Project Requirements

Enterprise Ransomware Assessment — Risk Register Mapped to NIST CSF 2.0

1. Software Requirements

The Python program uses only the Python standard library. No third-party packages are required.

Component	Requirement	Purpose
Python	Python 3.10+ recommended	Runs the CLI script and uses dataclasses/type hints.
Python libraries	Standard library only	argparse, csv, dataclasses, datetime, pathlib, typing.
Spreadsheet tool (optional)	Excel, LibreOffice Calc, or equivalent	Review and edit the generated CSV register.

2. Organizational Inputs

• Critical business services, applications, assets, data, and system dependencies.

• Threat scenarios, existing control evidence, and business impact context.

• Approved likelihood and impact scales and the organization's risk-rating thresholds.

• Named risk owners, treatment owners, target dates, and escalation / acceptance criteria.

• Relevant NIST CSF 2.0 Organizational Profile or implementation priorities, where available.

3. Generated Outputs

• CSV risk register containing risk ID, scenario, CSF function/category, likelihood, impact, inherent score, rating, controls, treatment, owner, target date, and status.

• Illustrative starter set of ten ransomware risks that should be reviewed and tailored.

• Score bands: 1–7 Low; 8–14 Moderate; 15–19 High; 20–25 Critical (example thresholds only).

Security and governance note

The project does not scan systems, execute malware, or certify compliance. Sample risks, scores, and mappings are illustrative. Validate them with organizational stakeholders and use your approved risk methodology before relying on the register.